



FET

32015021041 رهنف امجد غازي العدوان



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Question 18

Not yet answered

Marked out of 2.00

Flag question

Given that $f(x) = x^4 + 16x + 1$ then

$$\lim_{h \rightarrow 0} \frac{f(1+3h) - f(1)}{4h} =$$

Select one:

- ☐ a. 30
- ☐ b. 12
- ☐ c. 6
- ☒ d. 15

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Not yet answered

Marked out of 2.00

Flag question

If $y = f(x + y)$ and $f'(2) = 7$ then

$\left. \frac{dy}{dx} \right|_{\substack{x=1 \\ y=1}}$ is equal to:

Select one:

☐ a. $-\frac{6}{5}$

☒ b. $-\frac{7}{6}$

☐ c. $-\frac{5}{4}$

☐ d. $-\frac{4}{3}$

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Question 25

Not yet answered

Marked out of 2.00

Flag question

Evaluate the following limit:

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + x + 1} - 9x}{2x} =$$

Select one:

- ☐ a. $-\infty$
- ☒ b. -4
- ☐ c. 4
- ☐ d. ∞

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Question 21

Not yet answered

Marked out of 2.00

Flag question

Evaluate the following limit:

$$\lim_{x \rightarrow 2} \frac{4x-8}{2-x} =$$

Select one:

- ☒ a. -4
- ☐ b. does not exist
- ☐ c. 4
- ☐ d. 0

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Question 24

Not yet answered

Marked out of 2.00

Flag question

Evaluate:

$$\int_{-5}^5 \frac{x^3}{x^4 + \cos x} dx =$$

Select one:

- ☒ a. 0
- ☐ b. 10
- ☐ c. 5
- ☐ d. 15

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Question 14

Not yet answered

Marked out of 2.00

Flag question

Compute the following integral:

$$\int \sin 3x \cot 3x dx =$$

Select one:

- ☐ a. $\sin 3x + c$
- ☒ b. $-\frac{1}{3}\sin 3x + c$
- ☐ c. $-\sin 3x + c$
- ☐ d. $\frac{1}{3}\sin 3x + c$

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Question 7

Not yet answered

Marked out of 2.00

 Flag question

The range of the function

$$f(x) = \begin{cases} 0 & , x \leq -2 \\ \sqrt{4-x^2} & , -2 < x < 2 \\ 6 & , x \geq 2 \end{cases}$$

is:

Select one:

- ☐ a. $[0, 2) \cup \{6\}$
- ☐ b. $[0, 2]$
- ☒ c. $[0, 2] \cup \{6\}$
- ☐ d. $(0, 2)$

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Question 3

Not yet answered

Marked out of 1.00



Flag question

If $y = \sin x$ then $y^{(101)}\left(\frac{\pi}{6}\right) =$

Select one:

- ☒ a. $\frac{\sqrt{3}}{2}$
- ☐ b. 1
- ☐ c. $\frac{1}{2}$
- ☐ d. 0

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Question 22

Not yet answered

Marked out of 2.00

Flag question

Evaluate the following integral:

$$\int \frac{dx}{9x^2+12x+4} =$$

Select one:

- ☐ a. $\frac{1}{3x+2} + c$
- ☐ b. $\frac{1}{9x+6} + c$
- ☐ c. $\frac{-1}{9x+6} + c$
- ☒ d. $\frac{-1}{3x+2} + c$

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Question 5

Not yet answered

Marked out of 1.00

Flag question

The set of all the stationary points of the curve $y^3 + 3xy^2 - x^3 = 3$ is:

Select one:

- ☐ a. $\{(1, 1)\}$
- ☒ b. $\{(1, 1), (\sqrt[3]{3}, -\sqrt[3]{3})\}$
- ☐ c. $\{(\sqrt[3]{3}, -\sqrt[3]{3})\}$
- ☐ d. $\{(1, 1), (-\sqrt[3]{3}, \sqrt[3]{3})\}$

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Question 4

Not yet answered

Marked out of 1.00

 Flag question

The function $f(x) = \frac{1}{x-2}$ is:

Select one:

- ☐ a. increasing on $(2, \infty)$
- ☐ b. decreasing on $(-\infty, 2) \cup (2, \infty)$
- ☒ c. decreasing on the intervals $(-\infty, 2)$ and $(2, \infty)$
- ☐ d. increasing on $(-\infty, 2) \cup (2, \infty)$

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Not yet answered

Marked out of 1.00

Flag question

Given that $f(x) = \frac{x^2}{x^2-4}$ and

$$f'(x) = -\frac{8x}{(x^2-4)^2},$$

then the function f

Select one:

- ☒ a. is decreasing on the interval $(-2, 2)$
- ☐ b. is increasing on the interval $(-2, 2)$
- ☐ c. is concave up on the intervals $(-\infty, -2)$ and $(2, \infty)$
- ☐ d. is concave down on the intervals $(-\infty, -2)$ and $(2, \infty)$

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Question 3

Not yet answered

Marked out of 1.00

Flag question

At the point $(0, 0)$ the function $f(x) = \sqrt[3]{x^2}$ has

Select one:

- ☐ a. a vertical tangent
- ☐ b. a vertical asymptote
- ☐ c. a relative maximum
- ☒ d. a cusp

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Question 1

Not yet answered

Marked out of 1.00

Flag question

The range of the function;

$$f(x) = 2 \sec x$$

equals:

Select one:

- ☐ a. $\mathbb{R}$
- ☐ b. $\mathbb{R} - [-2, 2]$
- ☒ c. $\mathbb{R} - (-2, 2)$
- ☐ d. $(-2, 2)$

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Question 3

Not yet answered

Marked out of 2.00

Flag question

$$\lim_{x \rightarrow \infty} \frac{2x + \sin x}{x + \cos x} =$$

Select one:

☒ a. 2

☐ b. ∞

☐ c. 0

☐ d. 3

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Question 2

Not yet answered

Marked out of 1.00

Flag question

The minimum (absolute minimum) value of the function

$$f(x) = -\sin x + x \cos x + 2, x \in [-\pi, \pi]$$

is equal to:

Select one:

- ☐ a. $\pi + 2$
- ☐ b. $\pi + 1$
- ☒ c. $-\pi + 2$
- ☐ d. 0

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Question 11

Not yet answered

Marked out of 1.00

Flag question

If f is one-to-one function such that $f(1) = 3$
then $f^{-1}(3) + 5 =$

Select one:

- ☐ a. 4
- ☐ b. 1
- ☐ c. 8
- ☒ d. 6

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Question 9

Not yet answered

Marked out of 1.00

Flag question

The range of the function;
 $\{(1, 2), (2, 2), (3, 2)\}$ is:

Select one:

- ☐ a. $\{1, 2\}$
- ☐ b. $\{1, 2, 3\}$
- ☐ c. $\{1\}$
- ☒ d. $\{2\}$

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Question 6

Not yet answered

Marked out of 1.00

Flag question

$[x]$: Greatest integer $\leq x$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} 4[\sin 2x] =$$

Select one:

- ☐ a. 4
- ☒ b. -4
- ☐ c. 0
- ☐ d. does not exist

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Question 13

Not yet answered

Marked out of 1.00



Flag question

The natural domain of $f(x) = x^2 - 1$ is equal to:

Select one:

- ☐ a. $[1, \infty)$
- ☐ b. $[-1, 1]$
- ☐ c. $(-\infty, -1]$
- ☒ d. $\mathbb{R}$

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Question 16

Not yet answered

Marked out of 2.00

Flag question

$$\lim_{x \rightarrow \infty} \frac{\sqrt{4x^2 + x + 5x + 1}}{x + 10} =$$

Select one:

☒ a. 7

☐ b. 0

☐ c. 2

☐ d. 3

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Question 15

Not yet answered

Marked out of 2.00

Flag question

If n is an integer then

$$\lim_{x \rightarrow 0} \frac{\sin(20x + 2n\pi)}{4x} =$$

Select one:

- ☒ a. does not exist
- ☐ b. 5
- ☐ c. 0
- ☐ d. 4

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Question 7

Not yet answered

Marked out of 2.00

Flag question

If

$$f(x) = \begin{cases} \frac{\sin 2x}{3x} & \text{if } x \neq 0 \\ \frac{1}{k} & \text{if } x = 0 \end{cases}$$

then the value of k that makes f
continuous at $x = 0$ is equal to:

Select one:

☐ a. $\frac{2}{3}$

☒ b. $\frac{3}{2}$

☐ c. 1

☐ d. 0

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Question 18

Not yet answered

Marked out of 1.00

Flag question

Which of the following **functions** is one-to-one on its natural domain?

Select one:

- ☐ a. $f(x) = |x|$
- ☐ b. $f(x) = 4$
- ☒ c. $f(x) = x - 4$
- ☐ d. $f(x) = x^2 + 1$

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Question 20

Not yet answered

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Flag question

If $f(x) = 2 + 3 \sec x$ then
 $f\left(\frac{\pi}{3}\right) =$

Select one:

- ☒ a. 8
- ☐ b. 0
- ☐ c. 6
- ☐ d. 2

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Question 12

Not yet answered

Marked out of 1.00

Flag question

$$\lim_{x \rightarrow -4} (x^2 + x) =$$

Select one:

☐ a. -12

☐ b. 8

☒ c. 12

☐ d. -8

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Question 19

Not yet answered

Marked out of 2.00

Flag question

The least value of k that makes the function $f(x) = x^2 - 8x + 3$ to be one-to-one on the interval $[k, \infty)$ is equal to:

Select one:

- ☐ a. 3
- ☐ b. 5
- ☐ c. 6
- ☒ d. 4

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Question 17

Not yet answered

Marked out of 1.00



Flag question

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{7} =$$

Select one:

- ☐ a. $\frac{7}{2}$
- ☒ b. $\frac{2}{7}$
- ☐ c. does not exist
- ☐ d. 0

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Question 1

Not yet answered

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Flag question

If the tangent line to the curve of $y = 6 \tan 2x$ "when $x = \frac{\pi}{8}$ " intersects the

y -axis at the point A then the distance
(المسافة) between A and the origin
is equal to:

Select one:

☒ a. $4 - 2\pi$ ☐ b. $6 - 3\pi$ ☐ c. $2\pi - 4$ ☐ d. $3\pi - 6$

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Question 14

Not yet answered

Marked out of 2.00

Flag question

Given that $f(x) = x^3 - 5$ then
 $f^{-1}(x) =$

Select one:

☒ a. $\sqrt[3]{x} + 5$

☐ b. $\sqrt[3]{x - 5}$

☐ c. $\sqrt[3]{x} + 5$

☐ d. $\sqrt[3]{x} - 5$

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Question 5

Not yet answered

Marked out of 1.00

 Flag question

Given that

$$f(x) = x^{\frac{4}{3}} - 2x^{\frac{1}{3}} \quad \text{then} \quad f'(x) =$$

Select one:

☐ a. $\frac{4x-1}{\sqrt[3]{x^2}}$

☒ b. $\frac{4x-2}{3\sqrt[3]{x^2}}$

☐ c. $\frac{x-2}{3\sqrt[3]{x^2}}$

☐ d. $\frac{4x-1}{3\sqrt[3]{x^2}}$

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Question 4

Not yet answered

Marked out of 2.00

Flag question

The range of the function $f(x) = \frac{1}{\sqrt{x^2+4}}$ is equal to :

Select one:

- ☐ a. $(0, 2]$
- ☐ b. $[0, \frac{1}{2}]$
- ☒ c. $(0, \frac{1}{2}]$
- ☐ d. $[0, 2]$

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Question 2

Not yet answered

Marked out of 2.00

Flag question

If $y = 6$ and $x = 1$ are the horizontal & vertical asymptotes of the function;

$$f(x) = \frac{ax+4}{x-b}$$

then the order pair $(a, b) =$

Select one:

- ☐ a. (1, 6)
- ☐ b. (6, -1)
- ☒ c. (6, 1)
- ☐ d. (-1, 6)

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Question 4

Not yet answered

Marked out of 1.00

Flag question

If $f(x) = x^5 + 2x + 1$ then $f'(1) =$

Select one:

☒ a. 7

☐ b. 5

☐ c. 6

☐ d. 8

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Question 8

Not yet answered

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Flag question

If the graph of the function f is obtained by shifting the graph of

$$g(x) = |x| \text{ two units to the left, then } f(x) =$$

Select one:

☐ a. $|x| - 2$

☐ b. $|x - 2|$

☐ c. $|x| + 2$

☒ d. $|x + 2|$

Next page

**Question 4**

Not yet answered

Marked out of 2.00

Flag question

Given that $f(x) = \frac{x}{x-6}$ is a 1-1 function
then $f^{-1}(x) =$

Select one:

☐ a. $\frac{6x}{x+1}$

☐ b. $\frac{x}{x-6}$

☐ c. $\frac{x}{x+6}$

☒ d. $\frac{6x}{x-1}$

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Question 5

Not yet answered

Marked out of 1.00

Flag question

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x^3+1} - 5 \right) =$$

Select one:

- ☐ a. ∞
- ☐ b. -4
- ☐ c. 5
- ☒ d. -5

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Question 10

Not yet answered

Marked out of 1.00

Flag question

$$\lim_{x \rightarrow 64^-} \sqrt[3]{x} =$$

Select one:

☒ a. 4

☐ b. -64

☐ c. -4

☐ d. 64

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Question 5

Not yet answered

Marked out of 2.00

Flag question

If $y = \cos^4 2x - \sin^4 2x$ then $\frac{dy}{dx} =$

Select one:

- ☐ a. $4 \sin 4x$
- ☒ b. $-4 \sin 4x$
- ☐ c. $\sin 4x$
- ☐ d. $\sin 4x$

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Question 3

Not yet answered

Marked out of 2.00

Flag question

Achieve the following integral:

$$\int \csc x \cot x \, dx =$$

Select one:

- ☒ a. $-\csc x + c$
- ☐ b. $\csc x \tan x + c$
- ☐ c. $\csc x + c$
- ☐ d. $-\csc x \cot x + c$

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Question 1

Not yet answered

Marked out of 2.00

Flag question

Given that, $\lim_{x \rightarrow 1^+} f(x) = 10$,
 $\lim_{x \rightarrow 1^-} f(x) = 16$ and $f(1) = 4$

then $\lim_{x \rightarrow 5^-} f(6 - x) =$

Select one:

- ☐ a. 8
- ☐ b. 16
- ☒ c. 10
- ☐ d. does not exist

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Question 13

Not yet answered

Marked out of 2.00

Flag question

The **area** of the surface generated by revolving the line segment

$$y = \sqrt{3}x + 1, \quad 0 \leq x \leq 2$$

about the x-axis is given by :

Select one:

- ☐ a. $4\pi \int_0^2 (\sqrt{3}x + 2) dx$
- ☒ b. $4\pi \int_0^2 (\sqrt{3}x + 1) dx$
- ☐ c. $2\pi \int_0^2 (\sqrt{3}x + 1) dx$
- ☐ d. $2\pi \int_0^2 (\sqrt{3}x + 2) dx$

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Question 2

Not yet answered

Marked out of 2.00

Flag question

If $f(x) = \sqrt[3]{(x-2)^2}$ then $f''(x) =$

Select one:

☐ a. $-\frac{1}{9\sqrt[3]{(x-2)^4}}$

☐ b. $\frac{2}{9\sqrt[3]{(x-2)^4}}$

☒ c. $-\frac{2}{9\sqrt[3]{(x-2)^4}}$

☐ d. $\frac{1}{9\sqrt[3]{(x-2)^4}}$

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Not yet answered

Marked out of 2.00

Flag question

The natural domain of the function

$$f(x) = \frac{2\sqrt{x-2}}{\sqrt{x-2}}$$

is:

Select one:

- ☐ a. $\mathbb{R}$
- ☐ b. $[2, \infty)$
- ☒ c. $(2, \infty)$
- ☐ d. $\mathbb{R} \setminus \{2\}$

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Question 12

Not yet answered

Marked out of 2.00

Flag question

Evaluate the following limit:

$$\lim_{x \rightarrow \infty} x \sin\left(\frac{4}{x}\right) =$$

Select one:

- ☐ a. ∞
- ☐ b. $-\infty$
- ☐ c. 0
- ☒ d. 4

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Question 9

Not yet answered

Marked out of 2.00

Flag question

If f is concave up on $\mathbb{R}$ then one of the following statements must be true:

Select one:

- ☐ a. $f'(-1) < f'(-2)$
- ☒ b. $f'(2) < f'(3)$
- ☐ c. $f'(0) < f'(-1)$
- ☐ d. $f'(5) < f'(4)$

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**Question 19**

Not yet answered

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Flag question

Using the cylindrical shells method, the volume of the solid generated when the region bounded by

$$x = y^2 - 3y \text{ and the } y\text{-axis}$$

if it is revolved about the x-axis, is given by:

Select one:

☐ a. $\pi \int_0^3 (y^3 - 3y^2) dy$

☐ b. $\pi \int_0^3 (3y^2 - y^3) dy$

☒ c. $2\pi \int_0^3 (3y^2 - y^3) dy$

☐ d. $2\pi \int_0^3 (y^3 - 3y^2) dy$

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**Question 11**

Not yet answered

Marked out of 2.00

Flag question

If f is a continuous function then:

$$\int_0^8 \frac{f(x)}{f(x)+f(8-x)} dx =$$

Select one:

- ☐ a. 2
- ☐ b. 3
- ☐ c. 5
- ☒ d. 4

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Question 10

Not yet answered

Marked out of 2.00

Flag question

Evaluate the following integral:

$$\int 4x^3 \sec^2 x^4 dx =$$

Select one:

- ☐ a. $-\tan x^4 + c$
- ☐ b. $4 \tan x^4 + c$
- ☒ c. $\tan x^4 + c$
- ☐ d. $-4 \tan x^4 + c$

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Question 15

Not yet answered

Marked out of 2.00

Flag question

Given that $f'(6) = 3$, $g'(1) = 6$ and $g''(1) = 6$ then

$$(f \circ g')'(1) =$$

Select one:

- ☐ a. 15
- ☒ b. 18
- ☐ c. 30
- ☐ d. 12

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Question 8

Not yet answered

Marked out of 2.00

Flag question

The function $f(x) = \sqrt{x^2 - 9}$ is increasing on

Select one:

- ☐ a. $(-\infty, 0]$
- ☒ b. $[0, \infty)$
- ☐ c. $(-\infty, -3]$
- ☐ d. $[3, \infty)$

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Question 17

Not yet answered

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Flag question

 $[x] \leftrightarrow \text{greatest integer } \leq x$

Evaluate the following limit:

$$\lim_{x \rightarrow \infty} \frac{6[x]+1}{2x} =$$

Select one:

- ☐ a. 0
- ☐ b. 6
- ☐ c. ∞
- ☒ d. 3

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